

Table 1. Preservation of pairwise Pearson correlations of covariates in bootstrap virtual populations, referenced to NHANES 2021–2023.

Off-diagonal covariate pairs from the covariate set {BMI, BW, CrCL, RACE, SEX}. NHANES column: single-sample Pearson r ($n = 3576$). Cohort columns: pooled r across all subjects $\pm$ SD of r across 250 bootstrap datasets. Preservation footer: $|\Delta r| = |\text{pooled } r - \text{NHANES } r|$.

Covariate pair	NHANES ($n = 3576$)	N = 40 (250 datasets) ¹	N = 80 (250 datasets) ¹	N = 300 (250 datasets) ¹
Pairwise correlation ($r \pm \text{SD}$)				
BMI–BW	0.88	0.88 ± 0.03	0.88 ± 0.02	0.88 ± 0.01
BMI–CrCL	0.50	0.50 ± 0.07	0.50 ± 0.05	0.49 ± 0.03
BW–CrCL	0.58	0.59 ± 0.06	0.59 ± 0.04	0.58 ± 0.02
BMI–RACE	0.18	0.19 ± 0.07	0.18 ± 0.05	0.18 ± 0.02
BW–RACE	0.23	0.23 ± 0.06	0.24 ± 0.04	0.23 ± 0.02
CrCL–RACE	0.04	0.04 ± 0.07	0.04 ± 0.06	0.04 ± 0.03
BMI–SEX	-0.04	-0.05 ± 0.07	-0.04 ± 0.05	-0.05 ± 0.03
BW–SEX	0.29	0.28 ± 0.07	0.29 ± 0.05	0.29 ± 0.02
CrCL–SEX	0.08	0.08 ± 0.07	0.08 ± 0.05	0.08 ± 0.03
RACE–SEX ²	0.00	-0.03 ± 0.00	0.01 ± 0.00	0.00 ± 0.03
Preservation summary				
<i>Pairs within ± 0.05</i>	—	20/20	20/20	20/20
<i>Max Δr vs NHANES</i>	—	0.036	0.011	0.005
<i>Mean Δr vs NHANES</i>	—	0.010	0.004	0.002

¹ Stratified bootstrap by SEX x RACE with rejection gate $|\text{cor} - \text{NHANES}| \leq 0.05$ on the upper triangle; tolerance scaled by $\text{sqrt}(300 / N)$.

² SD = 0 for RACE–SEX at N = 40 and N = 80 reflects fixed per-cell counts in the SEX x RACE stratification.